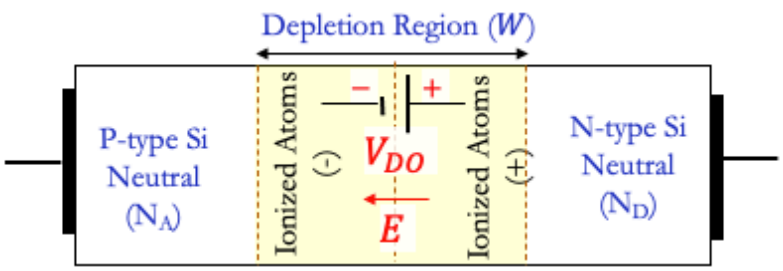
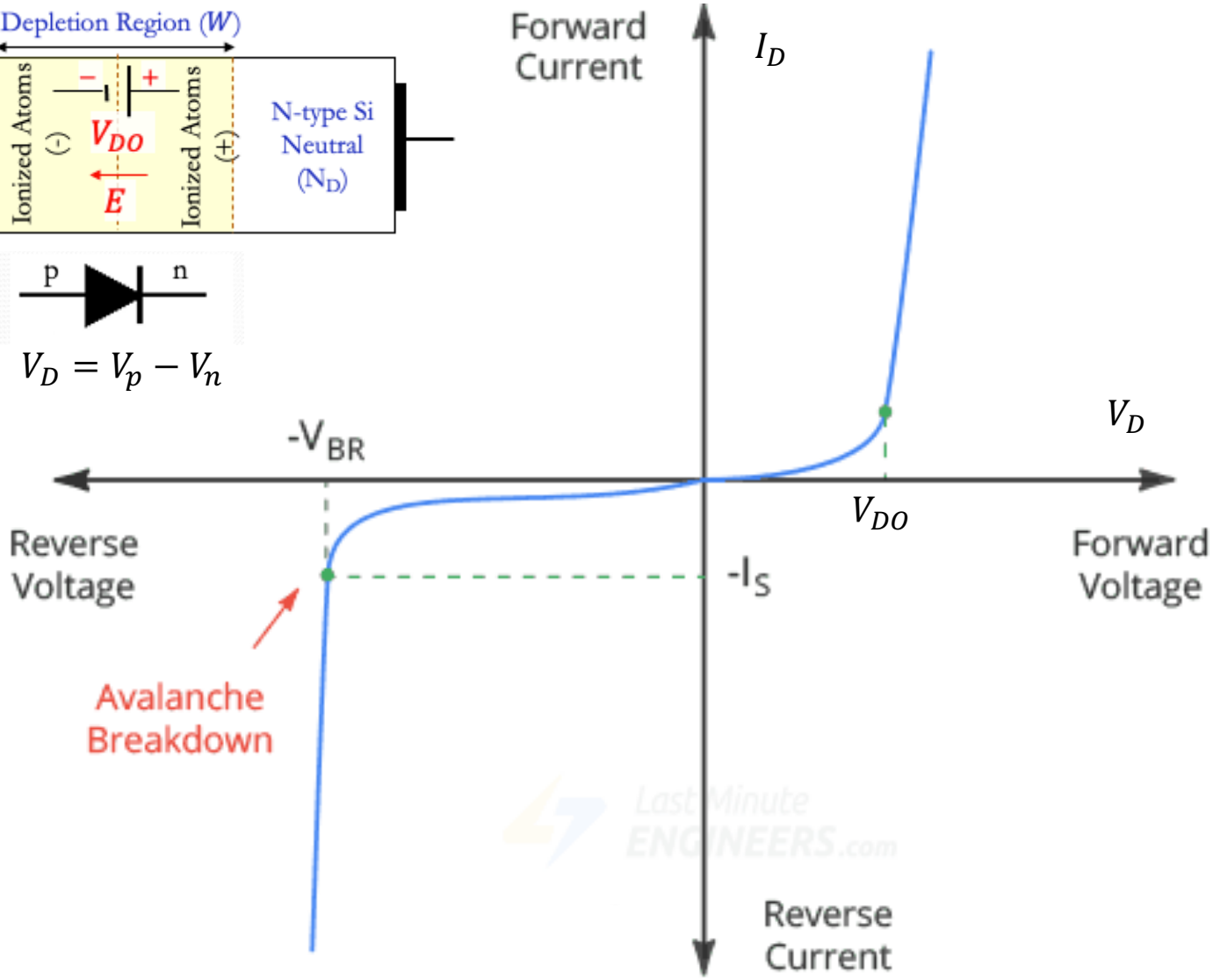
Analysis of Diode Circuits

- Diode Models.
- The exact model
- The ideal model

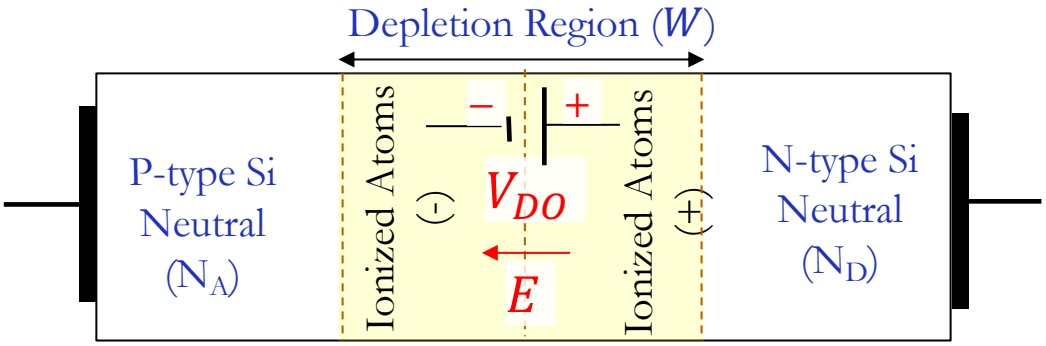
The PN junction: IV Curve

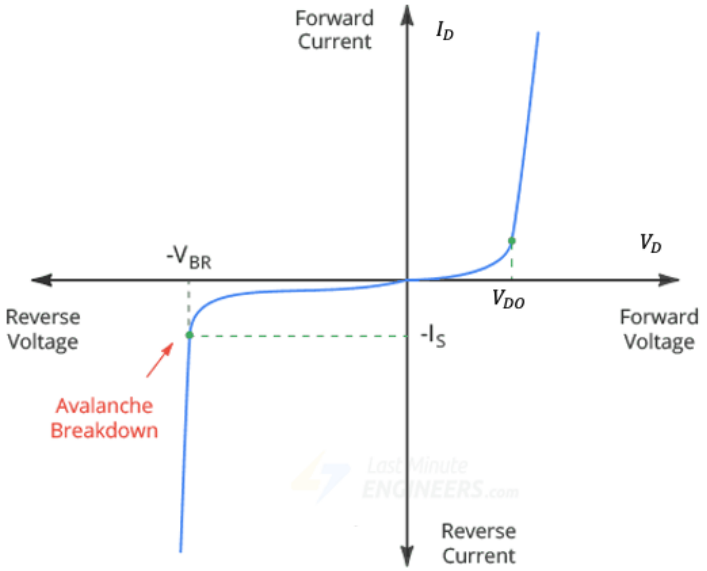


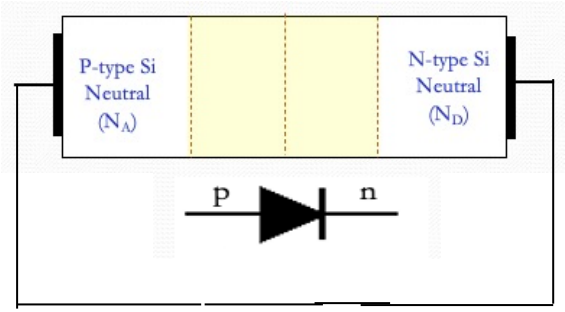
$$V_D = V_p - V_n$$



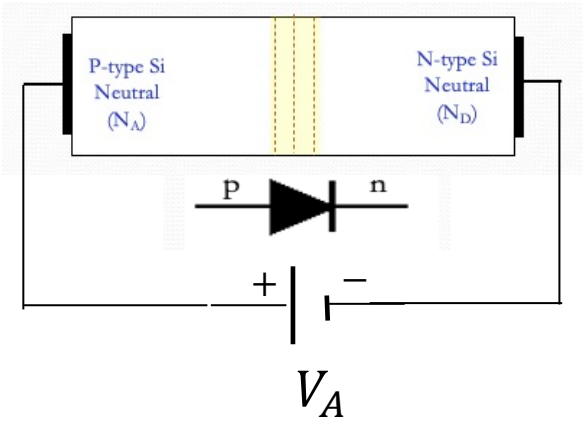
PN Junction Current and Biasing



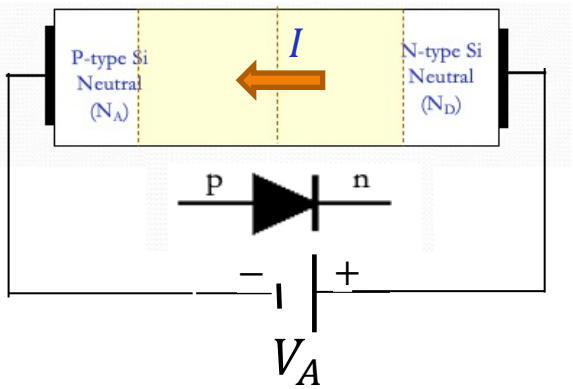




No Bias $I = 0$



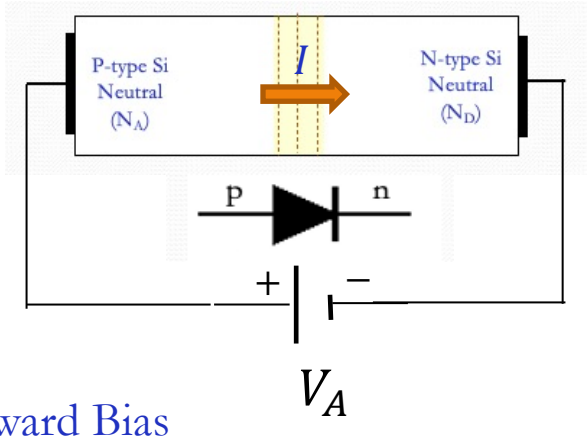
Forward Bias $I \neq 0$



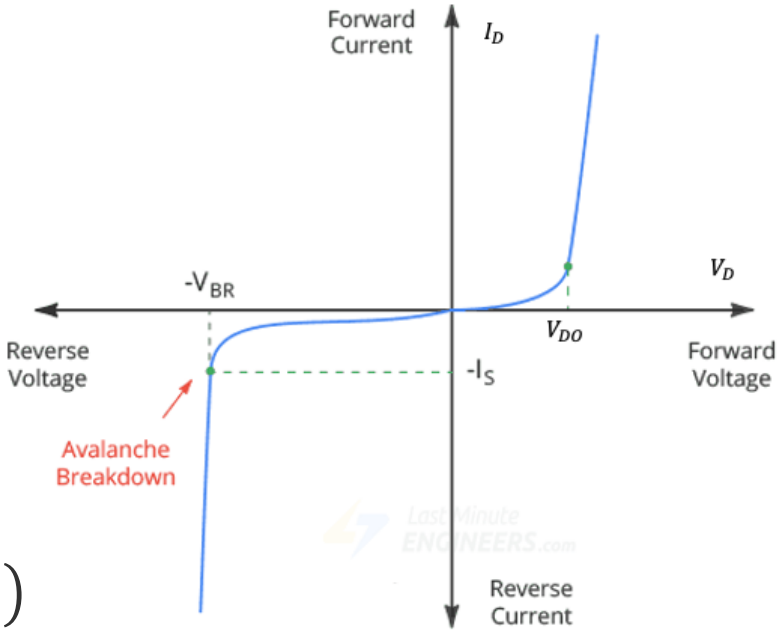
Reverse Bias $I = 0$

Breakdown $I \neq 0$

Current in Forward Region



Forward Bias



$$I_D = I_s\left(e^{V_D/nV_T} - 1\right)$$

Current through junction (A)

Saturation current (A)

Voltage across junction (V)
$V_D = V_p - V_n$

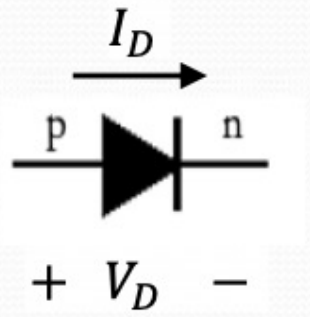
Ideality factor (1~2)

Thermal voltage (V)
(25.69 mV at room temp)

The Exact Diode Model

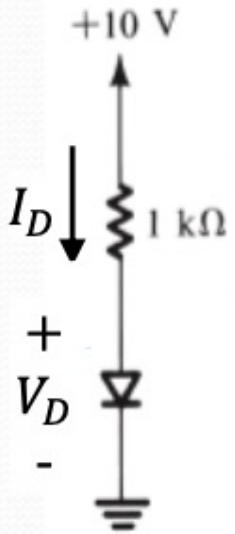
In this case we use the diode current equation

$$I_D = I_s\,(e^{V_D/nV_T} - 1)$$



Two unknowns, need another equation: the circuit equation.

Example



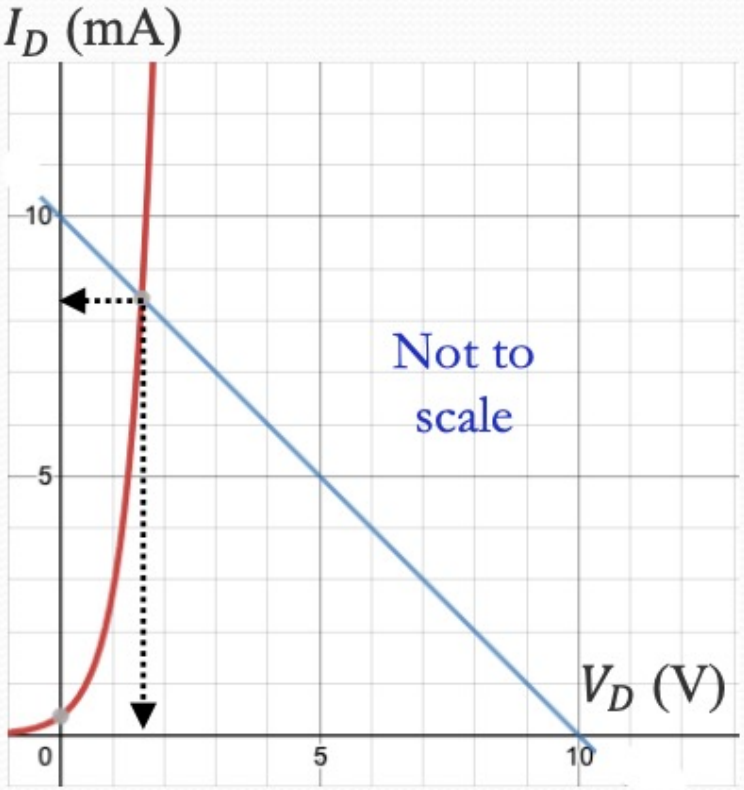
Diode equation

$$I_D = I_s\,(e^{V_D/nV_T} - 1)$$

Circuit equation

$$I_D = \frac{10 - V_D}{1\,k}$$

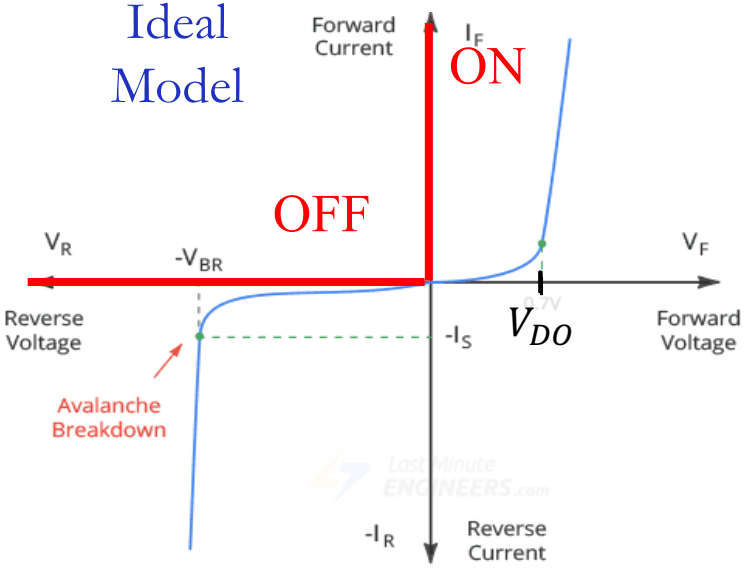
$I_s = 10^{-9}$ mA, $n = 1$



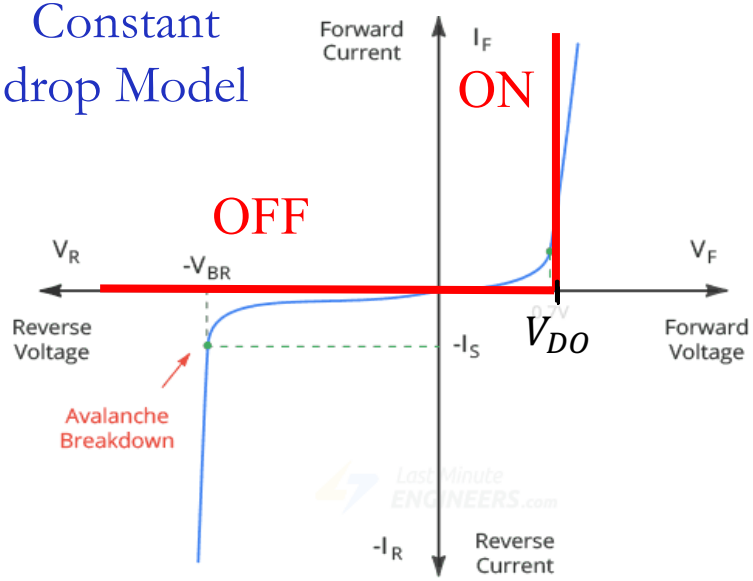
This nonlinear system of equations can be solved numerically using iterative methods (Newton's zero-finding method, etc.)

Simplified Diode Models

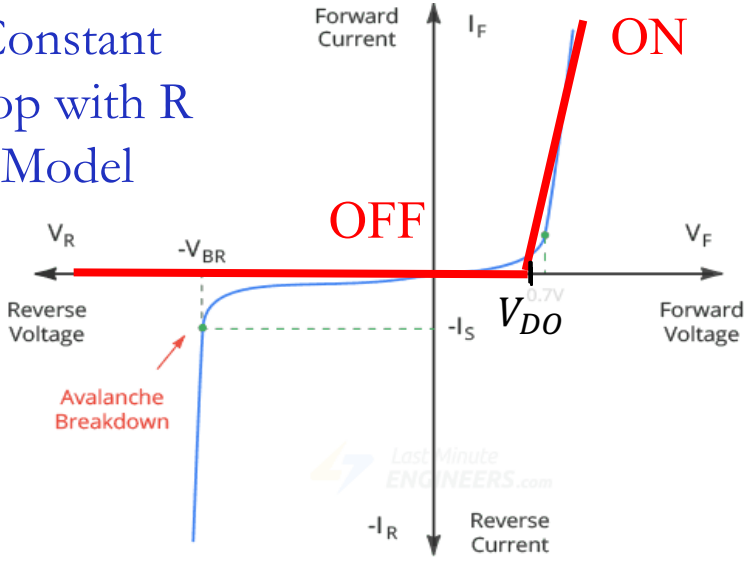
Ideal Model



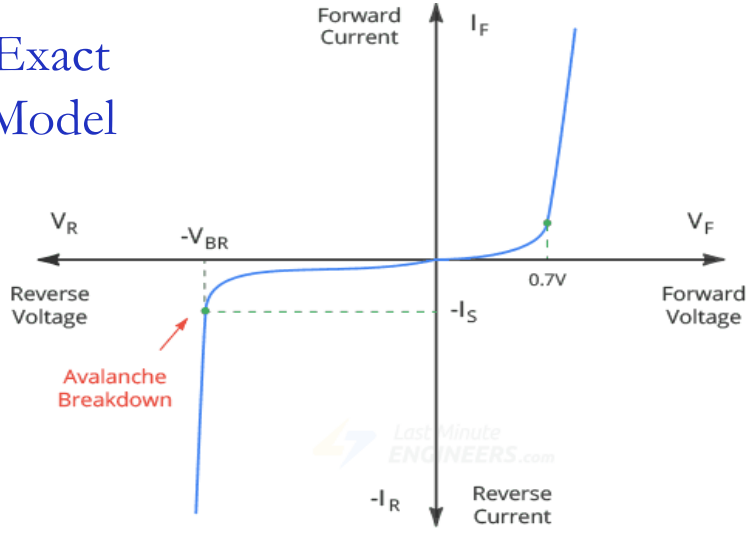
Constant drop Model



Constant drop with R Model

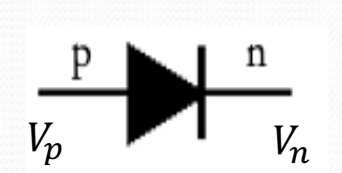


Exact Model

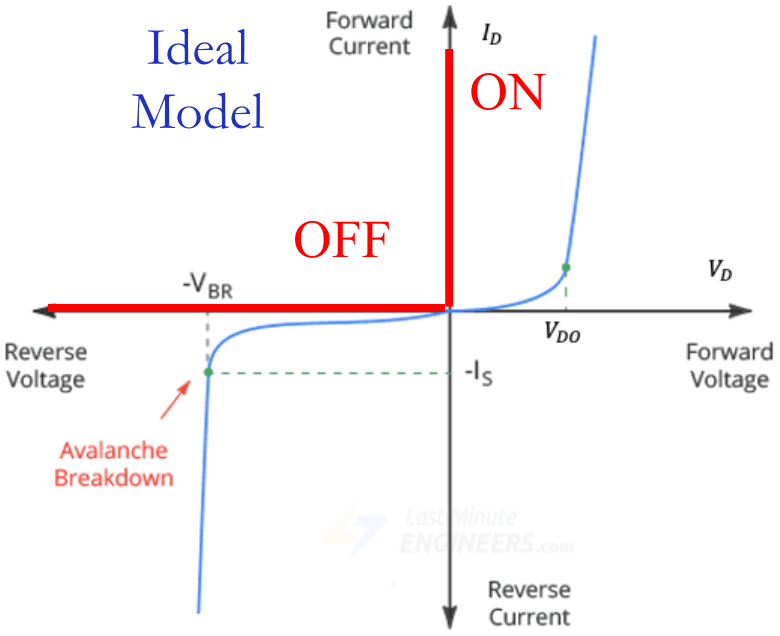


The Ideal Diode Model

$$V_D = V_p - V_n$$



Ideal Model



V_p V_n

P-side voltage $\geq$ N-side voltage: Diode ON

(Diode behaves like a short-circuit $I \neq 0, V_D = 0$)

V_p V_n

P-side voltage $<$ N-side voltage: Diode OFF

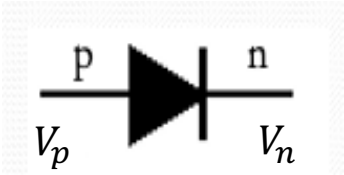
(Diode behaves like an open-circuit $I = 0$)

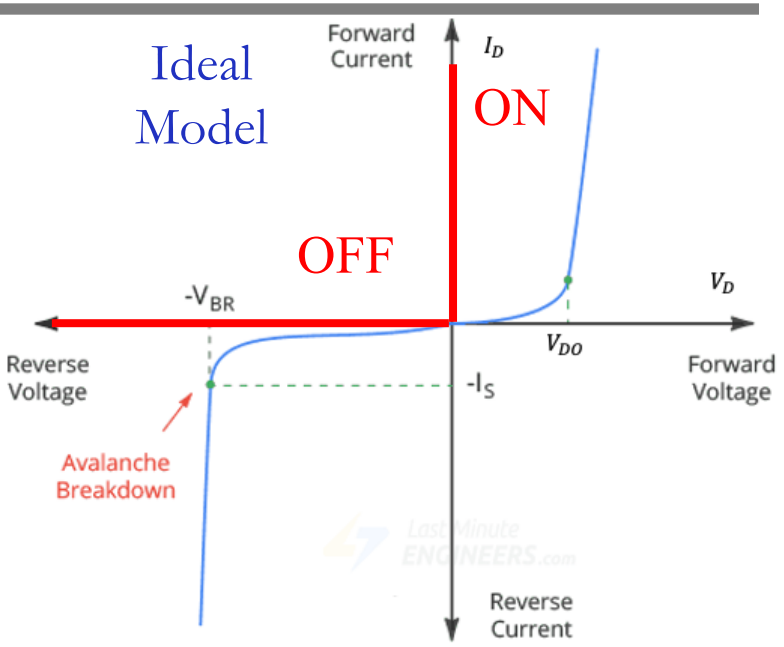
Advantages: Simple, quick answer.

Disadvantages: Not accurate

The Ideal Diode Model

$$V_D = V_p - V_n$$





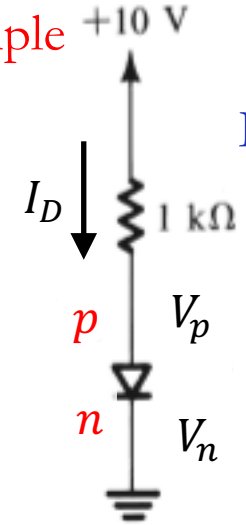
P-side voltage ≥ N-side voltage: Diode ON

(Diode behaves like a short-circuit $I \neq 0, V_D = 0$)

P-side voltage < N-side voltage: Diode OFF

(Diode behaves like an open-circuit $I = 0$)

Example



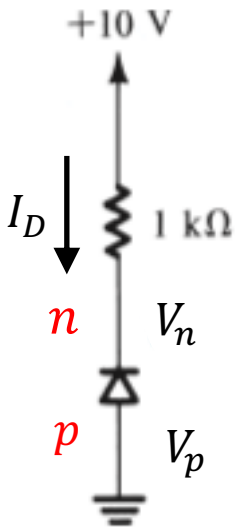
Looks like $V_p \geq V_n$

Diode ON (short circuit)

$$I_D = \frac{10 - 0}{1k}$$

$$I_D = 10\ mA$$

$$V_D = V_p - V_n = 0$$



Now $V_p < V_n$

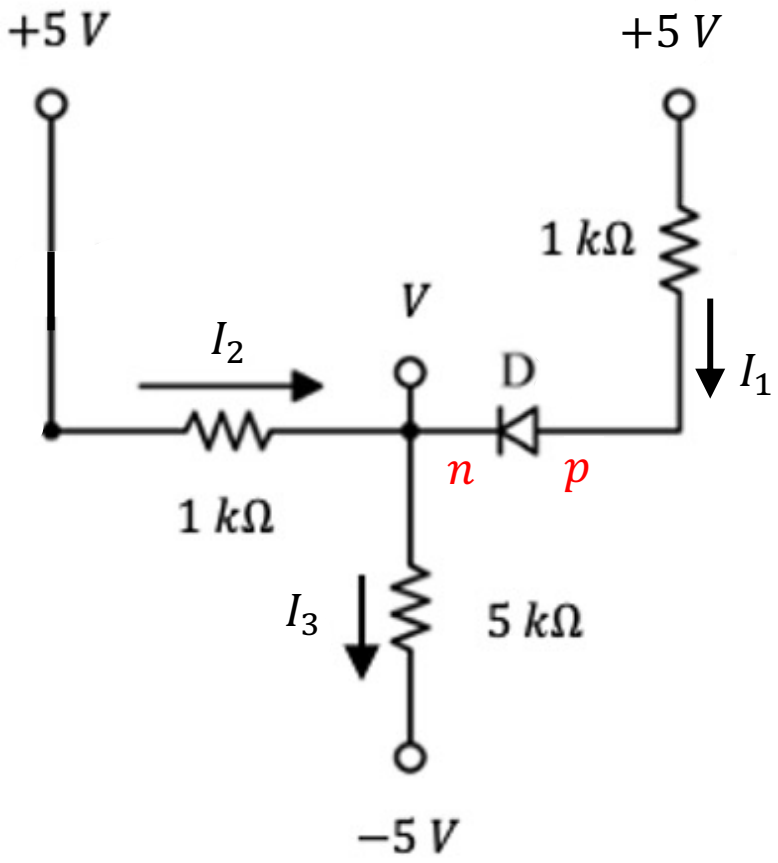
Diode OFF (open circuit)

$$I_D = 0$$

$$V_D = V_p - V_n = -10\ V$$

Examples

(1): Find the indicated currents and voltages. Use the ideal diode model.



D: Looks like $V_p \geq V_n$ Diode ON

Replace diode with short circuit.

Write KCL and KVL equations.

$$I_3 = I_1 + I_2 = 2\frac{5 - V}{1k} = \frac{V + 5}{5k}$$

$$50 - 10V = V + 5$$

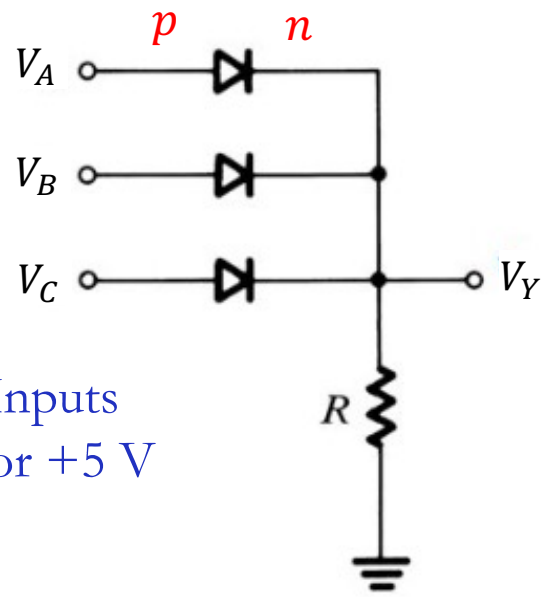
$$V = 4.09\ V$$

$$I_1 = I_2 = 0.91\ mA$$

$$I_3 = 1.82\ mA$$

Examples

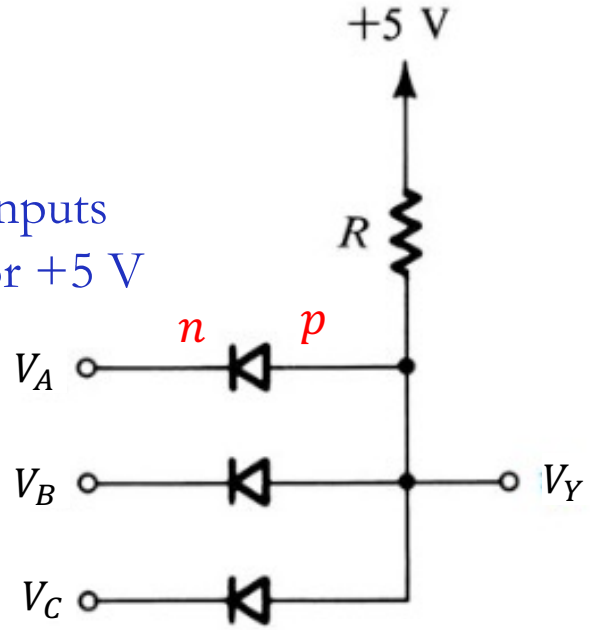
(2): Analyze the diode logic circuit below. Use the ideal diode model



Inputs
0 or +5 V

A	B	C	Y
0	0	0	0
1	0	0	1
1	1	0	1
1	1	1	1

OR gate



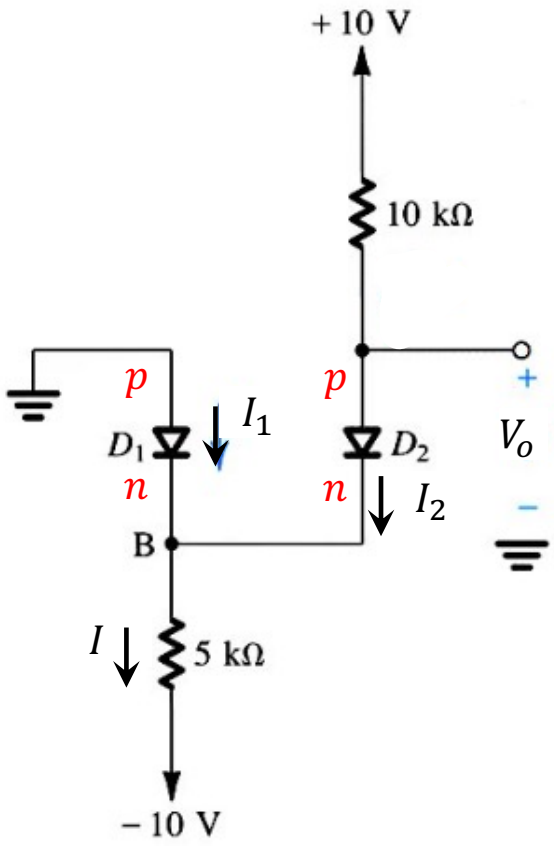
Inputs
0 or +5 V

A	B	C	Y
0	0	0	0
1	0	0	0
1	1	0	0
1	1	1	1

AND gate

Examples

(3): Analyze the diode circuit below. Use the ideal diode model



D1: Looks like $V_p \geq V_n$ Diode ON (short circuit)

D2: Looks like $V_p \geq V_n$ Diode ON (short circuit)

Point B at ground

$$I_2 = \frac{10 - 0}{10\ k} = 1\ mA$$

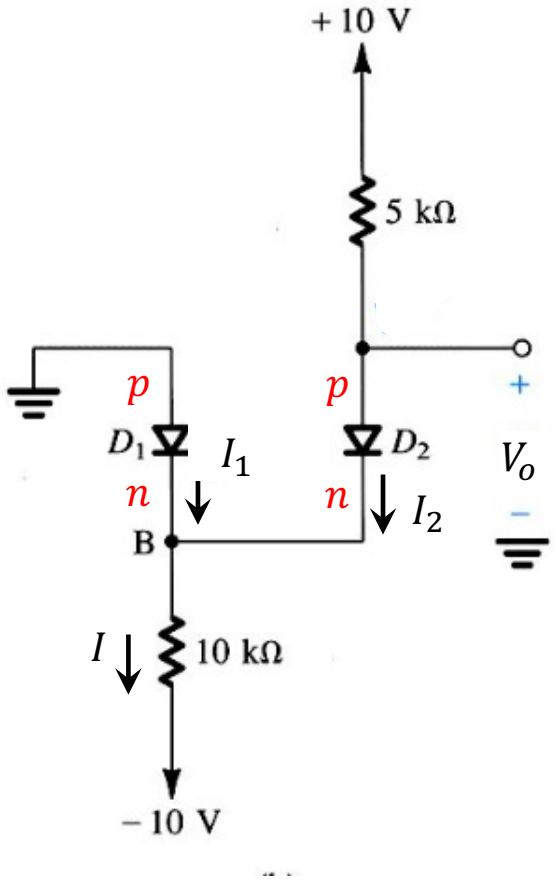
$$I = \frac{0 - (-10)}{5\ k} = 2\ mA$$

$$I_1 = I - I_2 = 1\ mA$$

$$V_o = 0$$

Examples

(4): Analyze the diode circuit below. Use the ideal diode model



D1: Looks like $V_p \geq V_n$ Diode ON

D2: Looks like $V_p \geq V_n$ Diode ON

Point B at ground $\quad I_2 = \dfrac{10-0}{5\ k} = 2\ mA$

$$I = \frac{0-(-10)}{10\ k} = 1\ mA \quad \Rightarrow \quad I_1 = I - I_{D2} = -1\ mA$$



Original assumption was wrong!!

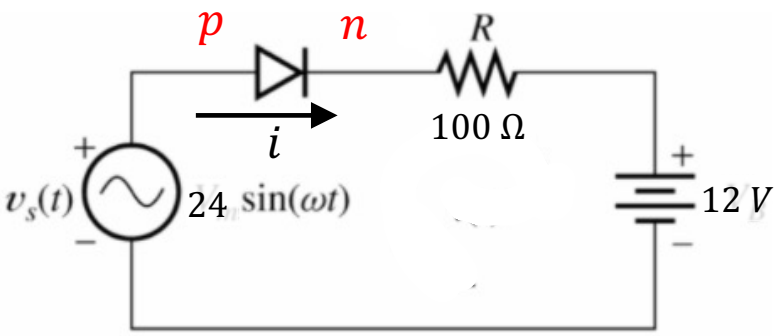
D1: OFF, D2: ON

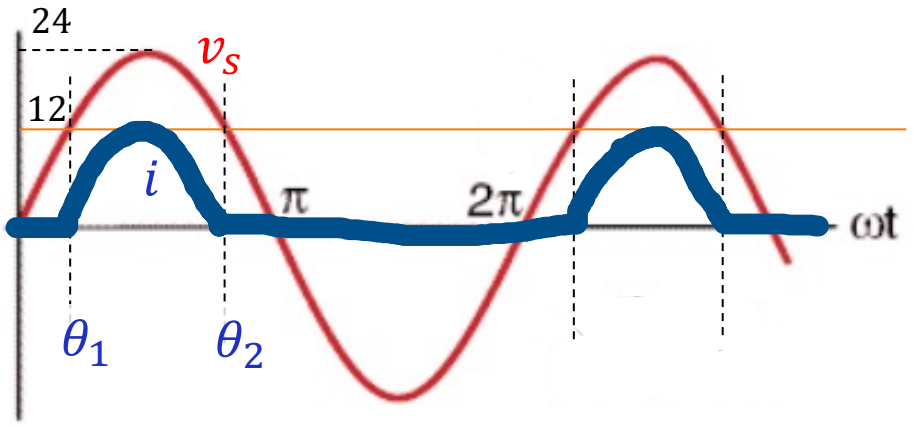
$$I_1 = 0 \qquad I_2 = I = \frac{10-(-10)}{10\ k + 5\ k} = 1.33\ mA$$

$$V_o = 10 - (5\ k)(1.33\ m) = 3.33\ V$$

Examples

(5): Find the current waveform. Use the ideal diode model





Diode is ON when $v_s \geq 12\ V$

→ Short circuit

$$i(t) = \frac{v_s - 12}{R} = \frac{24 \sin \omega t - 12}{100}$$

$$i_{D(max)} = \frac{24 - 12}{100} = 0.12\ A$$

Otherwise, Diode is OFF

Open circuit, $i = 0$

Conduction angle

$$12 = 24 \sin \theta_1 \quad \Rightarrow \quad \theta_1 = \sin^{-1} \frac{12}{24} = 30°$$

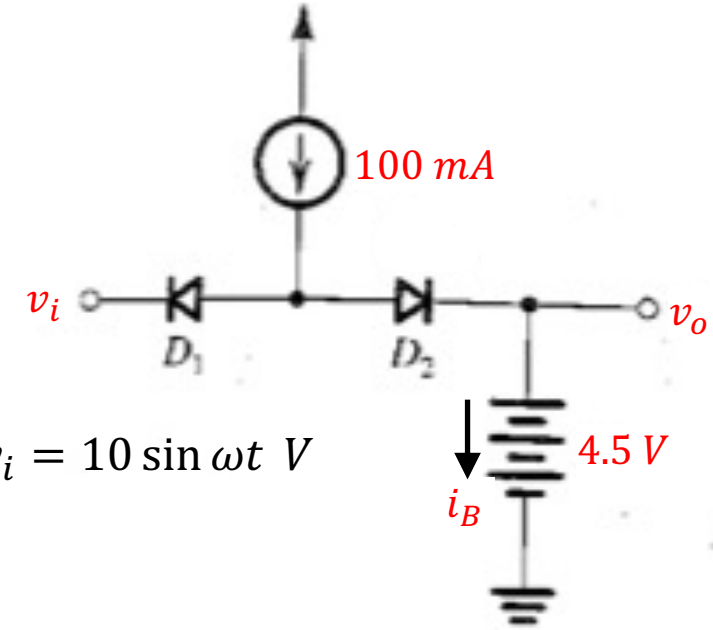
$$\Delta\theta = 180° - \theta_1 = 150°$$

Maximum (peak) reverse voltage

$$PIV = 24 + 12 = 36\ V$$

Examples

(6): In the battery-charging circuit, find the average battery current and angle of conduction. Use ideal model.



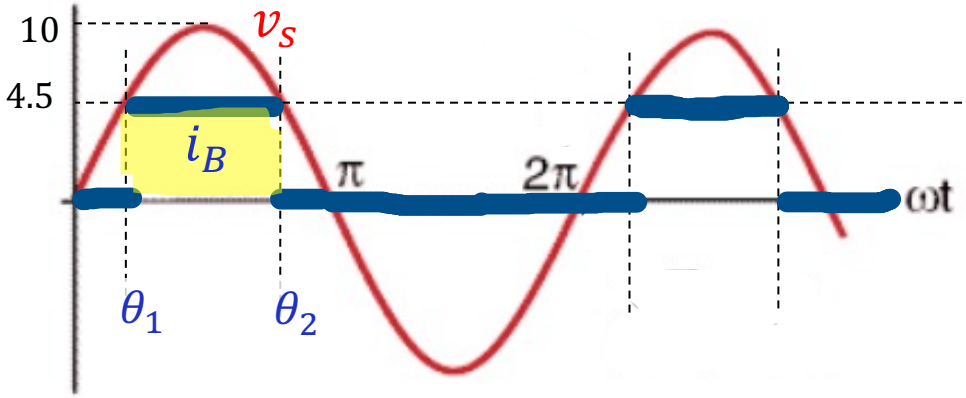
$v_i = 10 \sin \omega t \ V$

$v_i < 4.5\ V$: D1 ON D2 OFF

$i_B = 0$

$v_i > 4.5\ V$: D1 OFF D2 ON

$i_B = 100$ mA



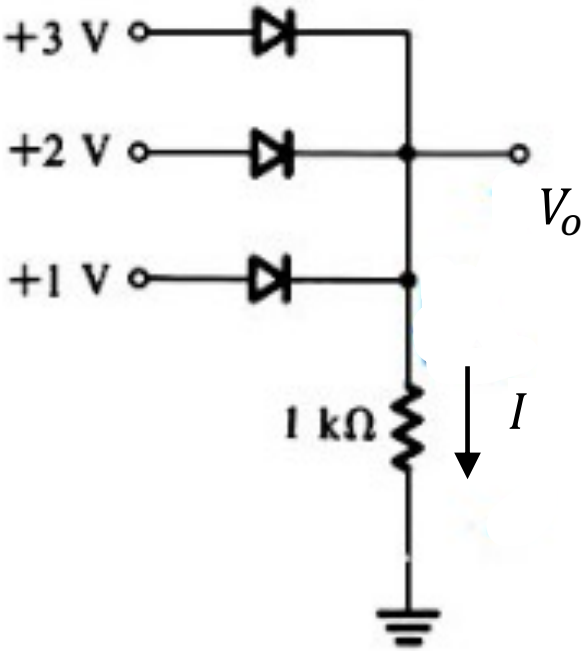
Conduction angle

$\theta_1 = 26.7°$ $\Delta\theta = 180° - 2\theta_1$

Average current?

Examples

(7): Analyze the diode circuit below. Use the ideal diode model



Q & A

Send me an email if
you have any
questions.